SMART LIBRARY MANAGEMENT PLATFORM

Mini-Project 4: DevOps and Cloud

- **Name: Shrawani Dipak Toge**
- **PRN:** 2125UDSF1082

Table of Contents

1. Abstract & Introduction
2. Problem Statement & Objectives
3. Scope & Technology Stack
4. System Architecture
5. Implementation Methodology
6. Implementation Phases & Evidence
7. Testing, Validation & Results
8. Challenges, Learnings & Conclusion
9. Future Enhancements

1. Abstract & Introduction

- **Abstract:** This project implements an automated deployment workflow for a **Smart Library Management Platform** using AWS cloud infrastructure, Terraform for Infrastructure as Code (IaC), Ansible for configuration management, and Docker/Nginx for containerized application delivery.

- **Introduction:** Leveraging cloud platforms and DevOps automation minimizes manual configuration overhead, ensures environmental consistency, and streamlines software delivery.

2. Problem Statement & Objectives

- **Problem Statement:** Manual server provisioning is time-consuming, prone to human error, and difficult to reproduce uniformly across environments.
- **Objectives:**
    - Provision AWS infrastructure (EC2 and S3 backup storage) via Terraform.
    - Automate server setup using Ansible.

- Containerize the web application using Docker and serve it via Nginx.
- Track source code changes using Git and GitHub.

3. Scope & Technology Stack

- **Scope:** Covers environment preparation, IaC deployment, server configuration, containerization, and accessibility verification.
- **Technology Stack:**

Technology	Purpose	Role in Project
AWS EC2	Cloud Compute	Hosts the web application server
AWS S3	Object Storage	Provides backup storage location
Terraform	IaC	Defines and provisions cloud resources
Ansible	Automation	Automates server setup and configuration
Docker	Containerization	Packages the application into an isolated container
Nginx	Web Server	Serves the web interface to clients
Git / GitHub	Version Control	Tracks and hosts project source code

4. System Architecture

The workflow executes sequentially from infrastructure provisioning to application delivery:

Terraform Configuration → AWS Resources (EC2 & S3) → Ansible Configuration → Docker + Nginx Container → Browser Access

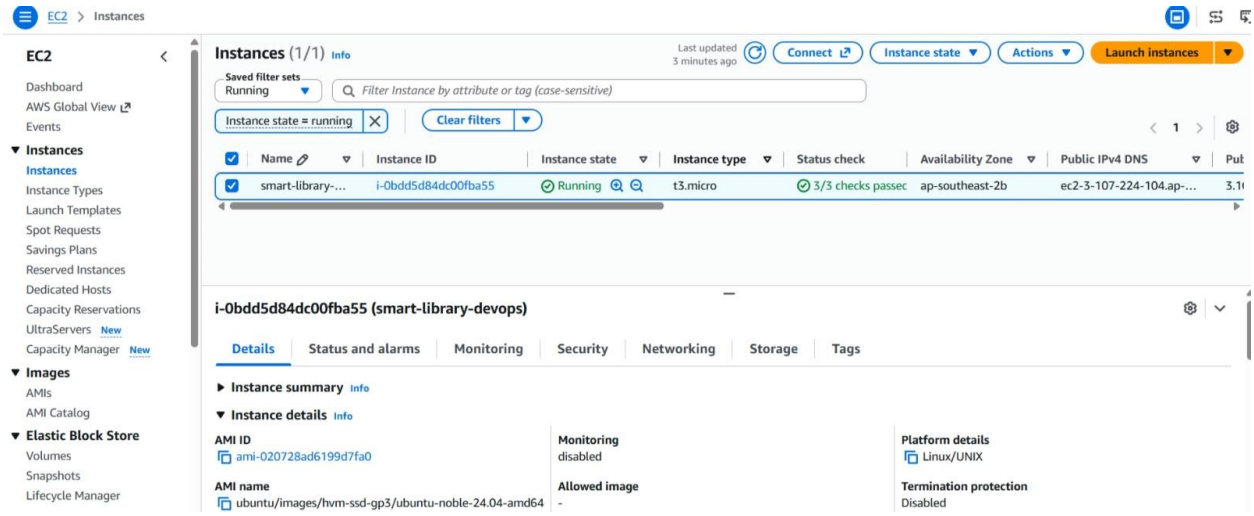
5. Implementation Methodology

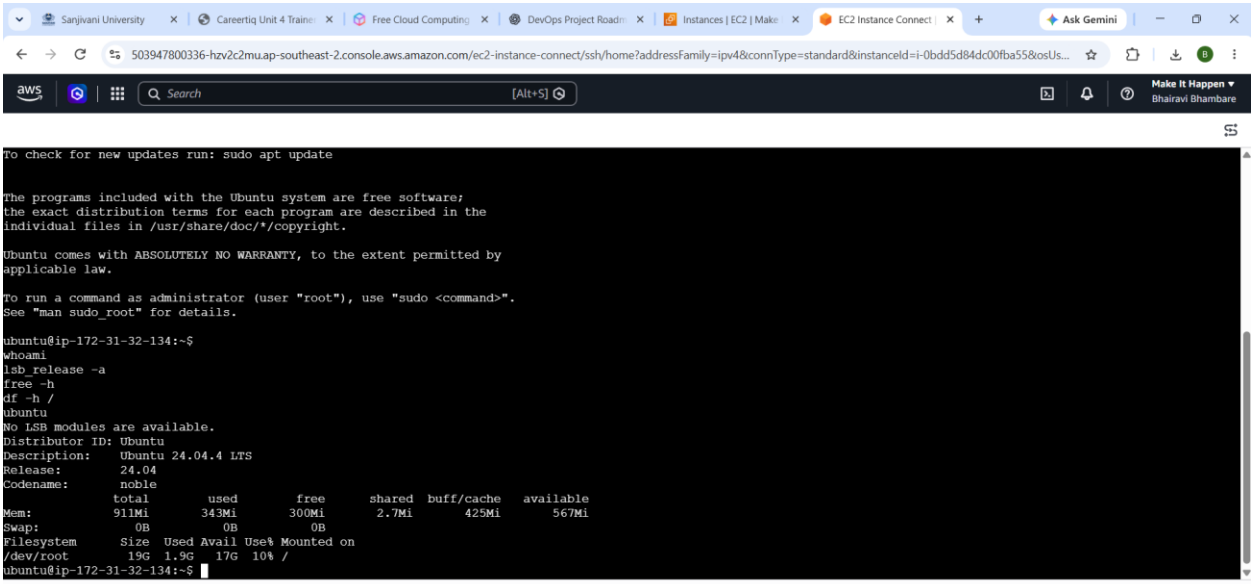
1. **Environment Setup:** Installed and verified tools (git, aws-cli, terraform, ansible, docker).

2. **Project Structure:** Organized local repository folders (terraform/, ansible/, app/, docs/).
3. **Containerization:** Built the Docker image containing the custom HTML application file (index.html).
4. **Infrastructure Provisioning:** Defined VPC, subnets, Internet Gateways, Route Tables, Security Groups, EC2, and S3 buckets via Terraform.
5. **Configuration Management:** Configured remote targets and deployed playbooks using Ansible.
6. **Verification:** Tested connectivity via curl and validated browser rendering over port 8080.

6. Implementation Phases & Evidence

- **6.1 Environment & Tools Setup:** Verified command-line utilities and AWS credentials
- **6.2 Project Structure & Application Source:** Initialized repository directories and created the HTML template
- **6.3 Docker Containerization:** Built smart-library-web:1.0 utilizing an Nginx base image
- **6.4 Terraform Infrastructure as Code:** Validated configuration files (main.tf) and generated execution plans
- **6.5 Server Configuration & Deployment:** Executed terraform apply to provision AWS resources and used Ansible playbooks for automated host configuration
- **6.6 Application Testing & Verification:** Checked playbook execution logs and verified active HTTP status codes
- **6.7 Security, Access Control & Version Control:** Configured security groups (SSH port 22, Web port 8080) and committed project changes to GitHub








```
ubuntu@ip-172-31-32-134:~$
aws --version
aws-cli/2.38.1 Python/3.14.6 Linux/6.17.0-1017-aws exe/x86_64.ubuntu.24
```

```
ubuntu@ip-172-31-32-134:~$ terraform -version
Terraform v1.16.6
on linux_amd64
```

```
ubuntu@ip-172-31-32-134:~$ ansible --version
ansible [core 2.16.3]
  config file = None
  configured module search path = ['/home/ubuntu/.ansible/plugins/modules', '/usr/share/ansible/plugins/modules']
  ansible python module location = /usr/lib/python3/dist-packages/ansible
  ansible collection location = /home/ubuntu/.ansible/collections:/usr/share/ansible/collections
  executable location = /usr/bin/ansible
  python version = 3.12.3 (main, Aug 31 2026, 10:18:26) [GCC 13.3.0] (/usr/bin/python3)
  jinja version = 3.1.2
  libyaml = True
```

```
ubuntu@ip-172-31-32-134:~$ git --version
git version 2.43.0
ubuntu@ip-172-31-32-134:~$ aws --version
aws-cli/2.38.1 Python/3.14.6 Linux/6.17.0-1017-aws exe/x86_64.ubuntu.24
ubuntu@ip-172-31-32-134:~$ docker --version
Command 'docker' not found, but can be installed with:
sudo apt install docker.io      # version 29.1.3-0ubuntu3~24.04.2, or
sudo apt install podman-docker  # version 4.9.3+ds1-1ubuntu0.2
ubuntu@ip-172-31-32-134:~$ aws sts get-caller-identity
{
    "UserId": "AROAXKVNBF4ILYOCWAD4J:i-0bdd5d84dc00fba55",
    "Account": "503947800336",
    "Arn": "arn:aws:sts::503947800336:assumed-role/smart-library-ec2-role/i-0bdd5d84dc00fba55"
}
```

```
ubuntu@ip-172-31-32-134:~$ docker --version
Docker version 29.1.3, build 29.1.3-0ubuntu3~24.04.2
ubuntu@ip-172-31-32-134:~$ sudo systemctl status docker --no-pager
● docker.service - Docker Application Container Engine
     Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: enabled)
     Active: active (running) since Mon 2026-10-19 16:13:49 UTC; 1min 26s ago
TriggeredBy: ● docker.socket
       Docs: https://docs.docker.com
   Main PID: 12588 (dockerd)
      Tasks: 9
     Memory: 85.0M (peak: 91.0M)
        CPU: 443ms
     CGroup: /system.slice/docker.service
             └─12588 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock
```

```
ubuntu@ip-172-31-32-134:~$ docker run hello-world
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
4f55086f7dd0: Pull complete
d5e71e642bf5: Download complete
Digest: sha256:11a3eb1597e864c51b3f56733cb2e8a90bf0c284bfc05f9444d81c558bc848aa
Status: Downloaded newer image for hello-world:latest

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
 1. The Docker client contacted the Docker daemon.
 2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
    (amd64)
 3. The Docker daemon created a new container from that image which runs the
    executable that produces the output you are currently reading.
 4. The Docker daemon streamed that output to the Docker client, which sent it
    to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
 $ docker run -it ubuntu bash
```

```
ubuntu@ip-172-31-32-134:~/smart-library-devops$ ls -la
find . -maxdepth 2 -type d | sort
cat README.md
total 32
drwxrwxr-x 6 ubuntu docker 4096 Oct 19 16:22 .
drwxr-x--- 8 ubuntu ubuntu 4096 Oct 19 16:20 ..
-rw-rw-r-- 1 ubuntu docker  108 Oct 19 16:22 .gitignore
-rw-rw-r-- 1 ubuntu docker  549 Oct 19 16:22 README.md
drwxrwxr-x 2 ubuntu docker 4096 Oct 19 16:20 ansible
drwxrwxr-x 2 ubuntu docker 4096 Oct 19 16:20 app
drwxrwxr-x 3 ubuntu docker 4096 Oct 19 16:20 docs
drwxrwxr-x 2 ubuntu docker 4096 Oct 19 16:20 terraform
.
./ansible
./app
./docs
./docs/screenshots
./terraform
# Smart Library Management Platform

## Project Overview
This project demonstrates automated AWS infrastructure provisioning
and web application deployment using Terraform, Ansible, Docker, and Git.

## Technology Stack
- AWS EC2, VPC, Security Groups, and S3
- Terraform
- Ansible
- Docker
- Git and GitHub
```

i-0bdd5d84dc00fba55 (smart-library-devops)

PublicIPs: 3.107.224.104 PrivateIPs: 172.31.32.134

```

## Project Structure
- terraform/: AWS infrastructure configuration
- ansible/: Server configuration and deployment automation
- app/: Web application
- docs/screenshots/: Project evidence and screenshots

## Deployment Status
In progress.
ubuntu@ip-172-31-32-134:~/smart-library-devops$
```

i-0bdd5d84dc00fba55 (smart-library-devops)

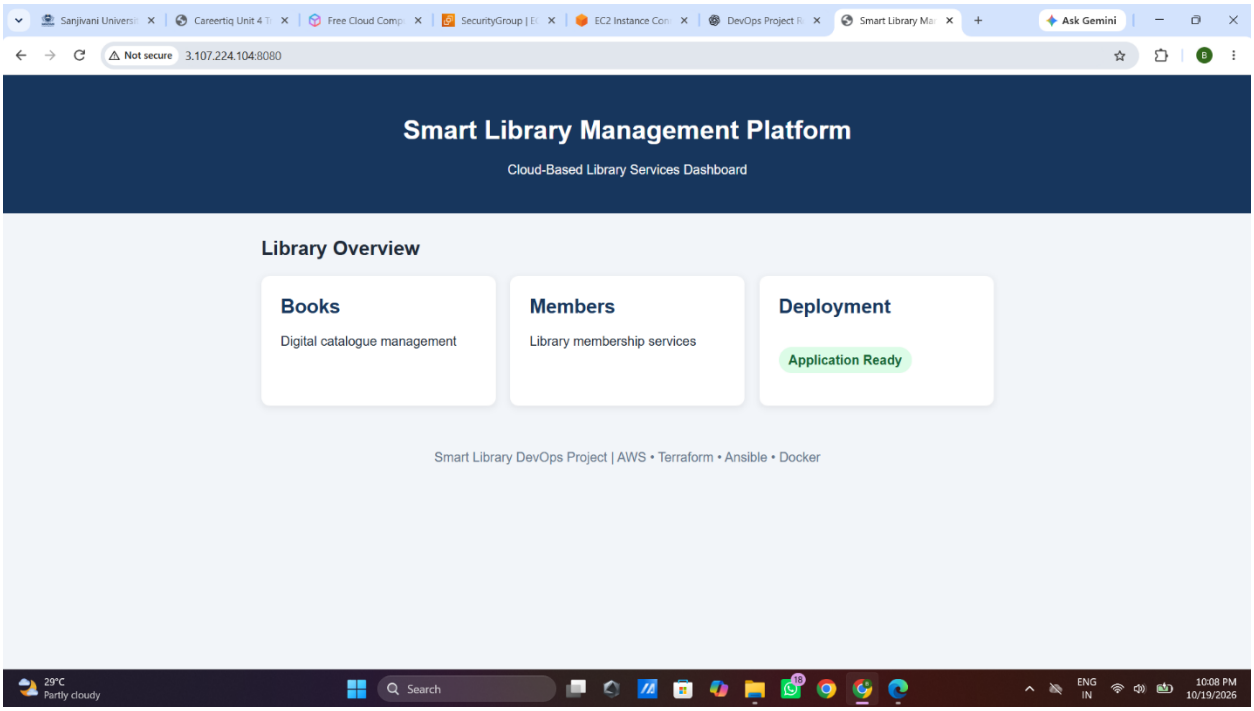
PublicIPs: 3.107.224.104 PrivateIPs: 172.31.32.134

```
ubuntu@ip-172-31-32-134:~/smart-library-devops$ ls -l app/
head -n 12 app/index.html
total 4
-rw-rw-r-- 1 ubuntu docker 1794 Oct 19 16:26 index.html
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Smart Library Management Platform</title>
  <style>
    body {
      margin: 0;
      font-family: Arial, sans-serif;
      background: #f2f5f9;
      color: #1f2937;
ubuntu@ip-172-31-32-134:~/smart-library-devops$
```

```
710ec83ca6cd: Download complete
70683d58fa5c: Download complete
e2de96513ba9: Pull complete
440345b60747: Download complete
3bf3671a4e6a: Pull complete
6711e0c4f5a7: Pull complete
b44956913298: Pull complete
6638b436592c: Pull complete
710ec83ca6cd: Pull complete
70683d58fa5c: Pull complete
688ea647499e: Download complete
1fdd0614a47e: Pull complete
Digest: sha256:892bb2152464f0c5d8972593a5f5afaf8054552580b699d95f8ff4af08c2d33d
Status: Downloaded newer image for nginx:alpine
 ---> 892bb2152464
Step 2/3 : COPY index.html /usr/share/nginx/html/index.html
 ---> 2269b7d9da74
Step 3/3 : EXPOSE 80
 ---> Running in aee1a65ebd24
 ---> Removed intermediate container aee1a65ebd24
 ---> 0b0bfd327c06
Successfully built 0b0bfd327c06
Successfully tagged smart-library-web:1.0
ubuntu@ip-172-31-32-134:~/smart-library-devops$ docker images
                                                                                                    i Info →   U  In Use
IMAGE                    ID             DISK USAGE   CONTENT SIZE   EXTRA
hello-world:latest       11a3eb1597e8       25.9kB         9.49kB    U
nginx:alpine             892bb2152464        103MB         29.7MB
smart-library-web:1.0    0b0bfd327c06        102MB         28.8MB
ubuntu@ip-172-31-32-134:~/smart-library-devops$
```

```
ubuntu@ip-172-31-32-134:~/smart-library-devops$ docker ps
CONTAINER ID   IMAGE                  COMMAND                  CREATED          STATUS          PORTS                                       NAMES
5c24d85a9700   smart-library-web:1.0   "/docker-entrypoint.…"   14 seconds ago   Up 13 seconds   0.0.0.0:8080->80/tcp, [::]:8080->80/tcp   smart-library-web
ubuntu@ip-172-31-32-134:~/smart-library-devops$ curl -I http://localhost:8080
HTTP/1.1 200 OK
Server: nginx/1.31.7
Date: Mon, 19 Oct 2026 16:33:30 GMT
Content-Type: text/html
Content-Length: 1794
Last-Modified: Mon, 19 Oct 2026 16:26:27 GMT
Connection: keep-alive
ETag: "6ad644b3-702"
Accept-Ranges: bytes

ubuntu@ip-172-31-32-134:~/smart-library-devops$ curl -s http://localhost:8080 | grep '<title>'
  <title>Smart Library Management Platform</title>
ubuntu@ip-172-31-32-134:~/smart-library-devops$
```




```
ubuntu@ip-172-31-32-134:~/smart-library-devops$ ls -l terraform/
total 4
-rw-rw-r-- 1 ubuntu docker 2161 Oct 19 16:42 main.tf
```

```
ubuntu@ip-172-31-32-134:~/smart-library-devops$ sed -n '1,240p' terraform/main.tf
terraform {
  required_version = ">= 1.5.0"

  required_providers {
    aws = {
      source  = "hashicorp/aws"
      version = "~> 5.0"
    }
  }
}

provider "aws" {
  region = "ap-southeast-2"
}

resource "aws_vpc" "library_vpc" {
  cidr_block           = "10.20.0.0/16"
  enable_dns_support   = true
  enable_dns_hostnames = true

  tags = {
    Name    = "smart-library-vpc"
    Project = "smart-library-devops"
  }
}

resource "aws_subnet" "library_subnet" {
  vpc_id                  = aws_vpc.library_vpc.id
  cidr_block              = "10.20.1.0/24"
```

```
resource "aws_subnet" "library_subnet" {
  vpc_id                  = aws_vpc.library_vpc.id
  cidr_block              = "10.20.1.0/24"
  map_public_ip_on_launch = true

  tags = {
    Name    = "smart-library-public-subnet"
    Project = "smart-library-devops"
  }
}

resource "aws_internet_gateway" "library_igw" {
  vpc_id = aws_vpc.library_vpc.id

  tags = {
    Name = "smart-library-igw"
  }
}

resource "aws_route_table" "library_route_table" {
  vpc_id = aws_vpc.library_vpc.id

  route {
    cidr_block = "0.0.0.0/0"
    gateway_id = aws_internet_gateway.library_igw.id
```

```
resource "aws_subnet" "library_subnet" {
  vpc_id                  = aws_vpc.library_vpc.id
  cidr_block              = "10.20.1.0/24"
  map_public_ip_on_launch = true

  tags = {
    Name    = "smart-library-public-subnet"
    Project = "smart-library-devops"
  }
}

resource "aws_internet_gateway" "library_igw" {
  vpc_id = aws_vpc.library_vpc.id

  tags = {
    Name = "smart-library-igw"
  }
}

resource "aws_route_table" "library_route_table" {
  vpc_id = aws_vpc.library_vpc.id

  route {
    cidr_block = "0.0.0.0/0"
    gateway_id = aws_internet_gateway.library_igw.id
```

```
  route {
    cidr_block = "0.0.0.0/0"
    gateway_id = aws_internet_gateway.library_igw.id
  }

  tags = {
    Name = "smart-library-public-rt"
  }
}

resource "aws_route_table_association" "library_route_assoc" {
  subnet_id      = aws_subnet.library_subnet.id
  route_table_id = aws_route_table.library_route_table.id
}

resource "aws_security_group" "library_sg" {
  name        = "smart-library-sg"
  description = "Allow SSH and web traffic for the Smart Library project"
  vpc_id      = aws_vpc.library_vpc.id

  ingress {
    description = "SSH - restrict to your own public IP before applying"
    from_port   = 22
    to_port     = 22
    protocol    = "tcp"
    cidr_blocks = ["0.0.0.0/0"]
```

```
  ingress {
    description = "SSH - restrict to your own public IP before applying"
    from_port   = 22
    to_port     = 22
    protocol    = "tcp"
    cidr_blocks = ["0.0.0.0/0"]
  }

  ingress {
    description = "Smart Library web application"
    from_port   = 8080
    to_port     = 8080
    protocol    = "tcp"
    cidr_blocks = ["0.0.0.0/0"]
  }

  egress {
    from_port   = 0
    to_port     = 0
    protocol    = "-1"
    cidr_blocks = ["0.0.0.0/0"]
  }

  tags = {
    Name    = "smart-library-sg"
    Project = "smart-library-devops"
  }
}
```

```
resource "aws_s3_bucket" "library_backup" {
  bucket_prefix = "smart-library-backup-"

  tags = {
    Name    = "smart-library-backup"
    Project = "smart-library-devops"
  }
}
```

```
ubuntu@ip-172-31-32-134:~/smart-library-devops$ cd ~/smart-library-devops/terraform
terraform init
Initializing the backend...

Initializing provider plugins...
- Finding hashicorp/aws versions matching "~> 5.0"...
- Installing hashicorp/aws v5.100.0...
- Installed hashicorp/aws v5.100.0 (signed by HashiCorp)

Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
ubuntu@ip-172-31-32-134:~/smart-library-devops/terraform$
```

```
ubuntu@ip-172-31-32-134:~/smart-library-devops/terraform$ terraform plan

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
  + create

Terraform will perform the following actions:

  # aws_internet_gateway.library_igw will be created
  + resource "aws_internet_gateway" "library_igw" {
      + arn      = (known after apply)
      + id       = (known after apply)
      + owner_id = (known after apply)
      + tags     = {
          + "Name" = "smart-library-igw"
        }
      + tags_all = {
          + "Name" = "smart-library-igw"
        }
      + vpc_id   = (known after apply)
    }

  # aws_route_table.library_route_table will be created
  + resource "aws_route_table" "library_route_table" {
      + arn             = (known after apply)
      + id              = (known after apply)
      + owner_id        = (known after apply)
      + propagating_vgws = (known after apply)
```

```
  # aws_route_table.library_route_table will be created
  + resource "aws_route_table" "library_route_table" {
      + arn              = (known after apply)
      + id               = (known after apply)
      + owner_id         = (known after apply)
      + propagating_vgws = (known after apply)
      + route            = [
          + {
              + cidr_block                = "0.0.0.0/0"
              + gateway_id                = (known after apply)
                # (11 unchanged attributes hidden)
            },
        ]
      + tags             = {
          + "Name" = "smart-library-public-rt"
        }
      + tags_all         = {
          + "Name" = "smart-library-public-rt"
        }
      + vpc_id           = (known after apply)
    }

  # aws_route_table_association.library_route_assoc will be created
  + resource "aws_route_table_association" "library_route_assoc" {
      + id             = (known after apply)
      + route_table_id = (known after apply)
      + subnet_id      = (known after apply)
    }
```

```
# aws_route_table_association.library_route_assoc will be created
+ resource "aws_route_table_association" "library_route_assoc" {
    + id             = (known after apply)
    + route_table_id = (known after apply)
    + subnet_id      = (known after apply)
  }

# aws_s3_bucket.library_backup will be created
+ resource "aws_s3_bucket" "library_backup" {
    + acceleration_status         = (known after apply)
    + acl                         = (known after apply)
    + arn                         = (known after apply)
    + bucket                      = (known after apply)
    + bucket_domain_name          = (known after apply)
    + bucket_prefix               = "smart-library-backup-"
    + bucket_regional_domain_name = (known after apply)
    + force_destroy               = false
    + hosted_zone_id              = (known after apply)
    + id                          = (known after apply)
    + object_lock_enabled         = (known after apply)
    + policy                      = (known after apply)
    + region                      = (known after apply)
    + request_payer               = (known after apply)
    + tags                        = {
        + "Name"    = "smart-library-backup"
        + "Project" = "smart-library-devops"
      }
    + tags_all                    = {
        + "Name"    = "smart-library-backup"
```

```
    + website_domain              = (known after apply)
    + website_endpoint            = (known after apply)

    + cors_rule (known after apply)

    + grant (known after apply)

    + lifecycle_rule (known after apply)

    + logging (known after apply)

    + object_lock_configuration (known after apply)

    + replication_configuration (known after apply)

    + server_side_encryption_configuration (known after apply)

    + versioning (known after apply)

    + website (known after apply)
  }

# aws_security_group.library_sg will be created
+ resource "aws_security_group" "library_sg" {
    + arn                    = (known after apply)
    + description            = "Allow SSH and web traffic for the Smart Library project"
    + egress                 = [
        + {
            + cidr_blocks      = [
                + "0.0.0.0/0",
```

```
            + "0.0.0.0/0",
          ]
        + from_port        = 0
        + ipv6_cidr_blocks = []
        + prefix_list_ids  = []
        + protocol         = "-1"
        + security_groups  = []
        + self             = false
        + to_port          = 0
          # (1 unchanged attribute hidden)
      },
  ]
+ id                     = (known after apply)
+ ingress                = [
    + {
        + cidr_blocks      = [
          + "0.0.0.0/0",
        ]
        + description      = "SSH - restrict to your own public IP before applying"
        + from_port        = 22
        + ipv6_cidr_blocks = []
        + prefix_list_ids  = []
        + protocol         = "tcp"
        + security_groups  = []
        + self             = false
        + to_port          = 22
      },
    + {
        + cidr_blocks      = [
          + "0.0.0.0/0",
```

```
          ]
        + description      = "SSH - restrict to your own public IP before applying"
        + from_port        = 22
        + ipv6_cidr_blocks = []
        + prefix_list_ids  = []
        + protocol         = "tcp"
        + security_groups  = []
        + self             = false
        + to_port          = 22
      },
    + {
        + cidr_blocks      = [
            + "0.0.0.0/0",
          ]
        + description      = "Smart Library web application"
        + from_port        = 8080
        + ipv6_cidr_blocks = []
        + prefix_list_ids  = []
        + protocol         = "tcp"
        + security_groups  = []
        + self             = false
        + to_port          = 8080
      },
  ]
+ name                   = "smart-library-sg"
+ name_prefix            = (known after apply)
+ owner_id               = (known after apply)
+ revoke_rules_on_delete = false
+ tags                   = {
    + "Name"    = "smart-library-sg"
```

```
          },
      ]
    + name                   = "smart-library-sg"
    + name_prefix            = (known after apply)
    + owner_id               = (known after apply)
    + revoke_rules_on_delete = false
    + tags                   = {
        + "Name"    = "smart-library-sg"
        + "Project" = "smart-library-devops"
      }
    + tags_all               = {
        + "Name"    = "smart-library-sg"
        + "Project" = "smart-library-devops"
      }
    + vpc_id                 = (known after apply)
  }

# aws_subnet.library_subnet will be created
+ resource "aws_subnet" "library_subnet" {
    + arn                                            = (known after apply)
    + assign_ipv6_address_on_creation                = false
    + availability_zone                              = (known after apply)
    + availability_zone_id                           = (known after apply)
    + cidr_block                                     = "10.20.1.0/24"
    + enable_dns64                                   = false
    + enable_resource_name_dns_a_record_on_launch    = false
    + enable_resource_name_dns_aaaa_record_on_launch = false
    + id                                             = (known after apply)
    + ipv6_cidr_block_association_id                 = (known after apply)
    + ipv6_native                                    = false
```

```
    + ipv6_native                                    = false
    + map_public_ip_on_launch                        = true
    + owner_id                                       = (known after apply)
    + private_dns_hostname_type_on_launch            = (known after apply)
    + tags                                           = {
        + "Name"    = "smart-library-public-subnet"
        + "Project" = "smart-library-devops"
      }
    + tags_all                                       = {
        + "Name"    = "smart-library-public-subnet"
        + "Project" = "smart-library-devops"
      }
    + vpc_id                                         = (known after apply)
  }

# aws_vpc.library_vpc will be created
+ resource "aws_vpc" "library_vpc" {
    + arn                                  = (known after apply)
    + cidr_block                           = "10.20.0.0/16"
    + default_network_acl_id               = (known after apply)
    + default_route_table_id               = (known after apply)
    + default_security_group_id            = (known after apply)
    + dhcp_options_id                      = (known after apply)
    + enable_dns_hostnames                 = true
    + enable_dns_support                   = true
    + enable_network_address_usage_metrics = (known after apply)
    + id                                   = (known after apply)
    + instance_tenancy                     = "default"
    + ipv6_association_id                  = (known after apply)
    + ipv6_cidr_block                      = (known after apply)
```

i-0bdd5d84dc00fba55 (smart-library-devops)

PublicIPs: 3.107.224.104 PrivateIPs: 172.31.32.134

```
      + default_network_acl_id               = (known after apply)
      + default_route_table_id               = (known after apply)
      + default_security_group_id            = (known after apply)
      + dhcp_options_id                      = (known after apply)
      + enable_dns_hostnames                 = true
      + enable_dns_support                   = true
      + enable_network_address_usage_metrics = (known after apply)
      + id                                   = (known after apply)
      + instance_tenancy                     = "default"
      + ipv6_association_id                  = (known after apply)
      + ipv6_cidr_block                      = (known after apply)
      + ipv6_cidr_block_network_border_group = (known after apply)
      + main_route_table_id                  = (known after apply)
      + owner_id                             = (known after apply)
      + tags                                 = {
          + "Name"    = "smart-library-vpc"
          + "Project" = "smart-library-devops"
        }
      + tags_all                             = {
          + "Name"    = "smart-library-vpc"
          + "Project" = "smart-library-devops"
        }
    }

Plan: 7 to add, 0 to change, 0 to destroy.


Note: You didn't use the -out option to save this plan, so Terraform can't guarantee to take exactly these actions if you run "terraform apply" now.
ubuntu@ip-172-31-32-134:~/smart-library-devops/terraform$
```

```
ubuntu@ip-172-31-32-134:~/smart-library-devops/terraform$ aws ec2 describe-instances \
  --instance-ids i-0bdd5d84dc00fba55 \
  --region ap-southeast-2 \
  --query 'Reservations[0].Instances[0].{State:State.Name,VpcId:VpcId,SubnetId:SubnetId,PublicIP:PublicIpAddress}' \
  --output table
-------------------------------------------------------------------------------
|                              DescribeInstances                              |
+---------------+-----------+---------------------------+--------------------------+
|   PublicIP    |  State    |         SubnetId          |          VpcId           |
+---------------+-----------+---------------------------+--------------------------+
|  3.107.224.104|  running  |  subnet-044aa7cdb968e2e67  |  vpc-0ff426164f085894f  |
+---------------+-----------+---------------------------+--------------------------+
ubuntu@ip-172-31-32-134:~/smart-library-devops/terraform$
```

```
ubuntu@ip-172-31-32-134:~/smart-library-devops/terraform$ aws ec2 describe-instances \
  --instance-ids i-0bdd5d84dc00fba55 \
  --region ap-southeast-2 \
  --query 'Reservations[0].Instances[0].SecurityGroups[*].{Name:GroupName,GroupId:GroupId}' \
  --output table
-------------------------------------------------
|              DescribeInstances              |
+------------------------+--------------------+
|        GroupId         |       Name         |
+------------------------+--------------------+
|  sg-01d01a8d36a3cd09e  |  launch-wizard-12  |
+------------------------+--------------------+
ubuntu@ip-172-31-32-134:~/smart-library-devops/terraform$
```

```
ubuntu@ip-172-31-32-134:~/smart-library-devops/terraform$ aws ec2 describe-key-pairs \
  --region ap-southeast-2 \
  --query 'KeyPairs[*].{KeyName:KeyName,KeyType:KeyType}' \
  --output table
--------------------------------------
|         DescribeKeyPairs           |
+-----------------------+------------+
|        KeyName        |  KeyType   |
+-----------------------+------------+
|  miniproject-key      |  rsa       |
|  k8s-key              |  rsa       |
|  python-docker-key    |  rsa       |
|  Docker-exercise-key  |  rsa       |
|  docker-exercise-key  |  rsa       |
|  exercise-docker-key  |  rsa       |
|  docker-python-key    |  rsa       |
|  key-pair             |  rsa       |
|  jenkins-CICD-Server  |  rsa       |
|  jenkins-key          |  rsa       |
|  miniproject4-key     |  rsa       |
|  jenkins-exercise-key |  rsa       |
|  smart-library-key    |  rsa       |
|  jenkins-cicd-key     |  rsa       |
+-----------------------+------------+
ubuntu@ip-172-31-32-134:~/smart-library-devops/terraform$
```

```
  filter {
    name   = "name"
    values = ["ubuntu/images/hvm-ssd-gp3/ubuntu-noble-24.04-amd64-server-*"]
  }

  filter {
    name   = "architecture"
    values = ["x86_64"]
  }

  filter {
    name   = "virtualization-type"
    values = ["hvm"]
  }
}

# EC2 host for the Smart Library project
resource "aws_instance" "library_host" {
  ami                         = data.aws_ami.ubuntu.id
  instance_type               = "t3.micro"
  subnet_id                   = aws_subnet.library_subnet.id
  vpc_security_group_ids      = [aws_security_group.library_sg.id]
  key_name                    = "miniproject-key"
EOFalue       = aws_s3_bucket.library_backup.bucketucket"C2 host"
ubuntu@ip-172-31-32-134:~/smart-library-devops/terraform$ terraform fmt
ubuntu@ip-172-31-32-134:~/smart-library-devops/terraform$ terraform validate
Success! The configuration is valid.

ubuntu@ip-172-31-32-134:~/smart-library-devops/terraform$
```

```
  + library_backup_bucket_name = (known after apply)
  + library_host_public_ip     = (known after apply)
ubuntu@ip-172-31-32-134:~/smart-library-devops/terraform$ terraform apply smart-library.tfplan
aws_vpc.library_vpc: Creating...
aws_s3_bucket.library_backup: Creating...
aws_s3_bucket.library_backup: Creation complete after 2s [id=smart-library-backup-20261019175238163000000001]
aws_vpc.library_vpc: Still creating... [00m10s elapsed]
aws_vpc.library_vpc: Creation complete after 12s [id=vpc-0ecb54f91a8724b98]
aws_internet_gateway.library_igw: Creating...
aws_subnet.library_subnet: Creating...
aws_security_group.library_sg: Creating...
aws_internet_gateway.library_igw: Creation complete after 0s [id=igw-09b581d78808781ee]
aws_route_table.library_route_table: Creating...
aws_route_table.library_route_table: Creation complete after 1s [id=rtb-0b7262ce6e96b5bf8]
aws_security_group.library_sg: Creation complete after 2s [id=sg-0563b479ad6135eb0]
aws_subnet.library_subnet: Still creating... [00m10s elapsed]
aws_subnet.library_subnet: Creation complete after 10s [id=subnet-020fc14ca7d2af935]
aws_route_table_association.library_route_assoc: Creating...
aws_instance.library_host: Creating...
aws_route_table_association.library_route_assoc: Creation complete after 0s [id=rtbassoc-02f6bcc5fcefbce92]
aws_instance.library_host: Still creating... [00m10s elapsed]
aws_instance.library_host: Creation complete after 12s [id=i-037be8bdf9362fabc]

Apply complete! Resources: 8 added, 0 changed, 0 destroyed.

Outputs:

library_backup_bucket_name = "smart-library-backup-20261019175238163000000001"
library_host_public_ip = "3.106.215.171"
ubuntu@ip-172-31-32-134:~/smart-library-devops/terraform$
```

```
ubuntu@ip-172-31-32-134:~/smart-library-devops/terraform$ terraform apply smart-library.tfplan
aws_vpc.library_vpc: Creating...
aws_s3_bucket.library_backup: Creating...
aws_s3_bucket.library_backup: Creation complete after 2s [id=smart-library-backup-20261019175238163000000001]
aws_vpc.library_vpc: Still creating... [00m10s elapsed]
aws_vpc.library_vpc: Creation complete after 12s [id=vpc-0ecb54f91a8724b98]
aws_internet_gateway.library_igw: Creating...
aws_subnet.library_subnet: Creating...
aws_security_group.library_sg: Creating...
aws_internet_gateway.library_igw: Creation complete after 0s [id=igw-09b581d78808781ee]
aws_route_table.library_route_table: Creating...
aws_route_table.library_route_table: Creation complete after 1s [id=rtb-0b7262ce6e96b5bf8]
aws_security_group.library_sg: Creation complete after 2s [id=sg-0563b479ad6135eb0]
aws_subnet.library_subnet: Still creating... [00m10s elapsed]
aws_subnet.library_subnet: Creation complete after 10s [id=subnet-020fc14ca7d2af935]
aws_route_table_association.library_route_assoc: Creating...
aws_instance.library_host: Creating...
aws_route_table_association.library_route_assoc: Creation complete after 0s [id=rtbassoc-02f6bcc5fcefbce92]
aws_instance.library_host: Still creating... [00m10s elapsed]
aws_instance.library_host: Creation complete after 12s [id=i-037be8bdf9362fabc]

Apply complete! Resources: 8 added, 0 changed, 0 destroyed.

Outputs:

library_backup_bucket_name = "smart-library-backup-20261019175238163000000001"
library_host_public_ip = "3.106.215.171"
ubuntu@ip-172-31-32-134:~/smart-library-devops/terraform$
```

```
ubuntu@ip-10-20-1-24:~$ whoami
hostname
lsb_release -a
ubuntu
ip-10-20-1-24
No LSB modules are available.
Distributor ID: Ubuntu
Description:    Ubuntu 24.04.5 LTS
Release:        24.04
Codename:       noble
ubuntu@ip-10-20-1-24:~$
```

```
ubuntu@ip-10-20-1-24:~$ docker --version
Docker version 29.1.3, build 29.1.3-0ubuntu3~24.04.2
```

```
ubuntu@ip-10-20-1-24:~$ sudo systemctl status docker --no-pager
● docker.service - Docker Application Container Engine
     Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: enabled)
     Active: active (running) since Tue 2026-10-20 05:39:50 UTC; 2min 10s ago
TriggeredBy: ● docker.socket
       Docs: https://docs.docker.com
   Main PID: 3391 (dockerd)
      Tasks: 9
     Memory: 27.8M (peak: 28.0M)
        CPU: 409ms
     CGroup: /system.slice/docker.service
             └─3391 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock

Oct 20 05:39:49 ip-10-20-1-24 dockerd[3391]: time="2026-10-20T05:39:49.916078555Z" level=info msg="Restoring containers: start."
Oct 20 05:39:49 ip-10-20-1-24 dockerd[3391]: time="2026-10-20T05:39:49.962098508Z" level=info msg="Deleting nftables IPv4 rules" error="exit status 1"
Oct 20 05:39:49 ip-10-20-1-24 dockerd[3391]: time="2026-10-20T05:39:49.968289871Z" level=info msg="Deleting nftables IPv6 rules" error="exit status 1"
Oct 20 05:39:50 ip-10-20-1-24 dockerd[3391]: time="2026-10-20T05:39:50.334330289Z" level=info msg="Loading containers: done."
Oct 20 05:39:50 ip-10-20-1-24 dockerd[3391]: time="2026-10-20T05:39:50.343325067Z" level=info msg="Docker daemon" commit="29.1.3-0ubuntu3~24.04.…sion=29.1.3
Oct 20 05:39:50 ip-10-20-1-24 dockerd[3391]: time="2026-10-20T05:39:50.343528368Z" level=info msg="Initializing buildkit"
Oct 20 05:39:50 ip-10-20-1-24 dockerd[3391]: time="2026-10-20T05:39:50.389255661Z" level=info msg="Completed buildkit initialization"
Oct 20 05:39:50 ip-10-20-1-24 dockerd[3391]: time="2026-10-20T05:39:50.396300138Z" level=info msg="Daemon has completed initialization"
Oct 20 05:39:50 ip-10-20-1-24 dockerd[3391]: time="2026-10-20T05:39:50.396741520Z" level=info msg="API listen on /run/docker.sock"
Oct 20 05:39:50 ip-10-20-1-24 systemd[1]: Started docker.service - Docker Application Container Engine.
Hint: Some lines were ellipsized, use -l to show in full.
ubuntu@ip-10-20-1-24:~$
```

```
ubuntu@ip-10-20-1-24:~$ sudo systemctl is-active docker
active
ubuntu@ip-10-20-1-24:~$
```

```
ubuntu@ip-10-20-1-24:~$ sudo docker run --rm hello-world
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
4f55086f7dd0: Pull complete
d5e71e642bf5: Download complete
Digest: sha256:11a3eb1597e864c51b3f56733cb2e8a90bf0c284bfc05f9444d81c558bc848aa
Status: Downloaded newer image for hello-world:latest

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
 1. The Docker client contacted the Docker daemon.
 2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
    (amd64)
 3. The Docker daemon created a new container from that image which runs the
    executable that produces the output you are currently reading.
 4. The Docker daemon streamed that output to the Docker client, which sent it
    to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
 $ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
 https://hub.docker.com/

For more examples and ideas, visit:
 https://docs.docker.com/get-started/
```

```
ubuntu@ip-172-31-32-134:~/smart-library-devops$ ansible --version
ansible [core 2.16.3]
  config file = None
  configured module search path = ['/home/ubuntu/.ansible/plugins/modules', '/usr/share/ansible/plugins/modules']
  ansible python module location = /usr/lib/python3/dist-packages/ansible
  ansible collection location = /home/ubuntu/.ansible/collections:/usr/share/ansible/collections
  executable location = /usr/bin/ansible
  python version = 3.12.3 (main, Aug 31 2026, 10:18:26) [GCC 13.3.0] (/usr/bin/python3)
  jinja version = 3.1.2
  libyaml = True
ubuntu@ip-172-31-32-134:~/smart-library-devops$
```

```
ubuntu@ip-172-31-32-134:~$ ls -l ~/.ssh/miniproject-key.pem
-r-------- 1 ubuntu ubuntu 1678 Oct 20 06:16 /home/ubuntu/.ssh/miniproject-key.pem
ubuntu@ip-172-31-32-134:~$
```

```
ubuntu@ip-172-31-32-134:~/smart-library-devops$ ansible -i ansible/inventory.ini web -m ping
The authenticity of host '3.106.215.171 (3.106.215.171)' can't be established.
ED25519 key fingerprint is SHA256:OxSsX9un6Ecy60aSVgxIXynmEa17KG43yHINa4E+nac.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
3.106.215.171 | SUCCESS => {
    "changed": false,
    "ping": "pong"
}
ubuntu@ip-172-31-32-134:~/smart-library-devops$
```

```
ubuntu@ip-172-31-32-134:~/smart-library-devops$ ansible-galaxy collection list community.docker

# /usr/lib/python3/dist-packages/ansible_collections
Collection       Version
---------------- -------
community.docker 3.7.0
ubuntu@ip-172-31-32-134:~/smart-library-devops$
```

```
ubuntu@ip-172-31-32-134:~/smart-library-devops$ ansible-playbook -i ansible/inventory.ini ansible/deploy-library.yml --syntax-check

playbook: ansible/deploy-library.yml
ubuntu@ip-172-31-32-134:~/smart-library-devops$ ansible-playbook -i ansible/inventory.ini ansible/deploy-library.yml --list-tasks

playbook: ansible/deploy-library.yml

  play #1 (web): Deploy Smart Library web application   TAGS: []
    tasks:
      Install Docker    TAGS: []
      Ensure Docker is running  TAGS: []
      Create application directory      TAGS: []
      Copy Smart Library HTML page      TAGS: []
      Run Smart Library container       TAGS: []
ubuntu@ip-172-31-32-134:~/smart-library-devops$
```

```
ubuntu@ip-172-31-32-134:~/smart-library-devops$ ansible-playbook -i ansible/inventory.ini ansible/deploy-library.yml

PLAY [Deploy Smart Library web application] ************************************************************

TASK [Gathering Facts] *********************************************************************************
ok: [3.106.215.171]

TASK [Install Docker] **********************************************************************************
ok: [3.106.215.171]

TASK [Ensure Docker is running] ************************************************************************
ok: [3.106.215.171]

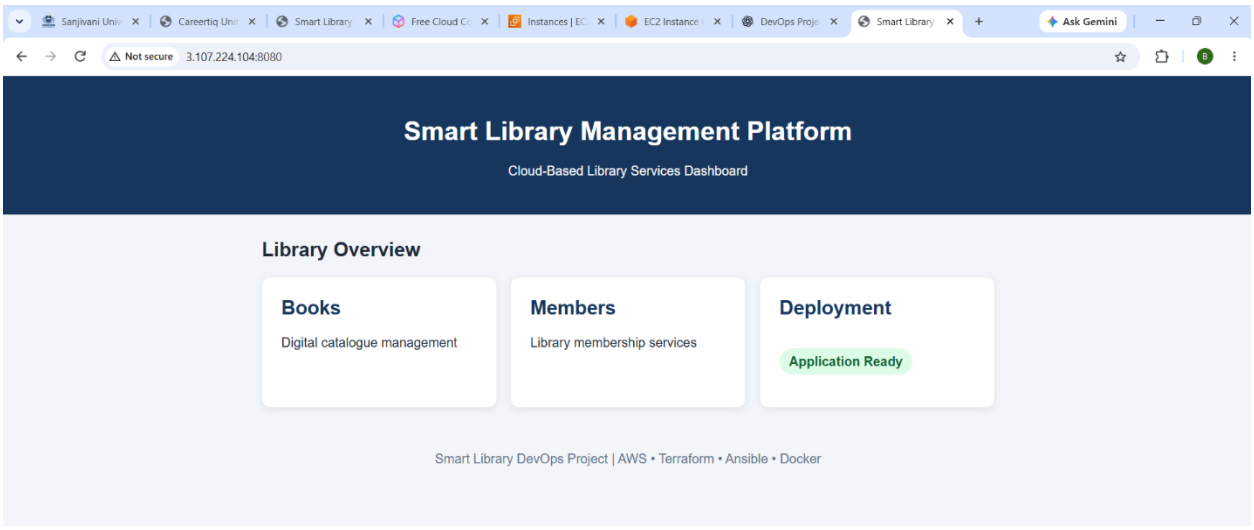
TASK [Create application directory] ********************************************************************
changed: [3.106.215.171]

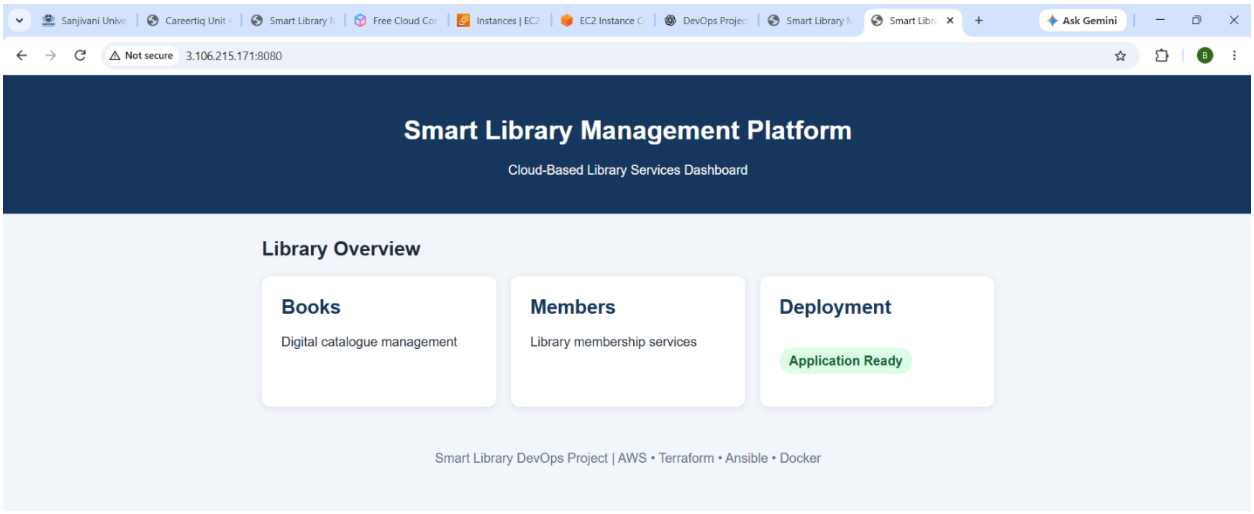
TASK [Copy Smart Library HTML page] ********************************************************************
changed: [3.106.215.171]

TASK [Run Smart Library container] *********************************************************************
changed: [3.106.215.171]

PLAY RECAP *********************************************************************************************
3.106.215.171              : ok=6    changed=3    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

ubuntu@ip-172-31-32-134:~/smart-library-devops$
```






```
ubuntu@ip-172-31-32-134:~/smart-library-devops$ ls -R
.:
README.md  ansible  app  docs  terraform

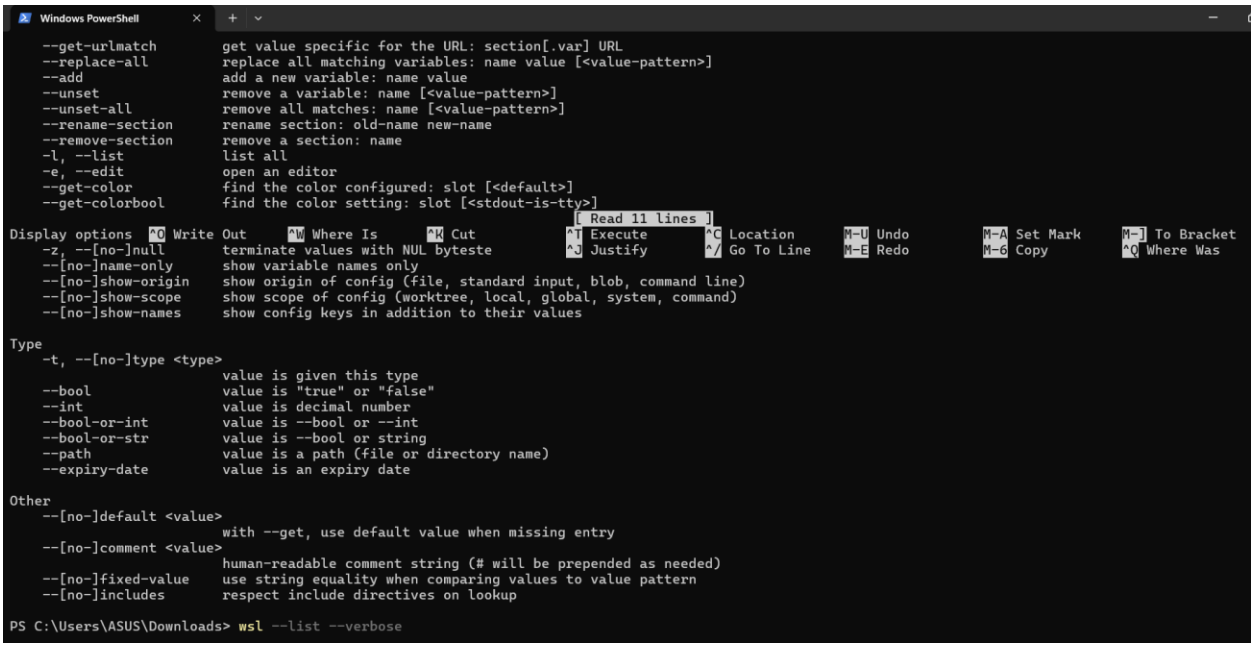
./ansible:
deploy-library.yml  inventory.ini

./app:
Dockerfile  index.html

./docs:
screenshots

./docs/screenshots:

./terraform:
main.tf  main.tf.backup  smart-library.tfplan  terraform.tfstate
ubuntu@ip-172-31-32-134:~/smart-library-devops$
```




```
ubuntu@ip-172-31-32-134:~/smart-library-devops$ git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   .gitignore
        new file:   README.md
        new file:   ansible/deploy-library.yml
        new file:   ansible/inventory.ini.example
        new file:   app/Dockerfile
        new file:   app/index.html
        new file:   terraform/.terraform.lock.hcl
        new file:   terraform/main.tf

ubuntu@ip-172-31-32-134:~/smart-library-devops$
```

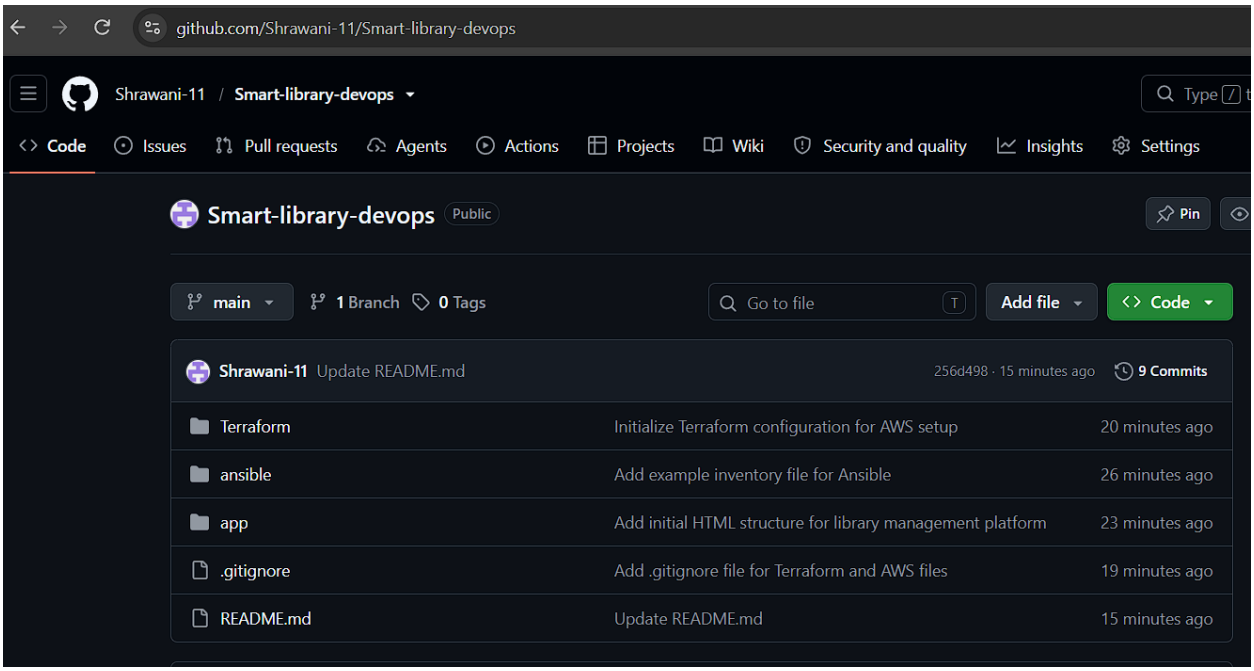
```
git diff --cached --check
ubuntu@ip-172-31-32-134:~/smart-library-devops$ git commit -m "Initial Smart Library DevOps project"
[master (root-commit) 0098477] Initial Smart Library DevOps project
 Committer: Ubuntu <ubuntu@ip-172-31-32-134.ap-southeast-2.compute.internal>
Your name and email address were configured automatically based
on your username and hostname. Please check that they are accurate.
You can suppress this message by setting them explicitly. Run the
following command and follow the instructions in your editor to edit
your configuration file:

    git config --global --edit

After doing this, you may fix the identity used for this commit with:

    git commit --amend --reset-author

 8 files changed, 339 insertions(+)
 create mode 100644 .gitignore
 create mode 100644 README.md
 create mode 100644 ansible/deploy-library.yml
 create mode 100644 ansible/inventory.ini.example
 create mode 100644 app/Dockerfile
 create mode 100644 app/index.html
 create mode 100644 terraform/.terraform.lock.hcl
 create mode 100644 terraform/main.tf
ubuntu@ip-172-31-32-134:~/smart-library-devops$ git log -1 --oneline
0098477 (HEAD -> master) Initial Smart Library DevOps project
ubuntu@ip-172-31-32-134:~/smart-library-devops$
```



7. Testing, Validation & Results

- **Syntax & Tool Checks:** Validated Terraform configuration files successfully using terraform validate.
- **Connectivity Testing:** Verified successful HTTP 200 responses and rendered UI dashboards via the EC2 public IP address on port 8080.

8. Challenges, Learnings & Conclusion

- **Challenges:** Managing dependencies across multiple DevOps tools (Terraform, Ansible, Docker) and ensuring proper network security group port mapping.
- **Learnings:** Emphasizes the significance of automated repeatable workflows, early configuration validation, and structured infrastructure pipelines.
- **Conclusion:** Successfully established a reproducible automated cloud deployment pipeline for the Smart Library Management Platform using AWS and DevOps practices.